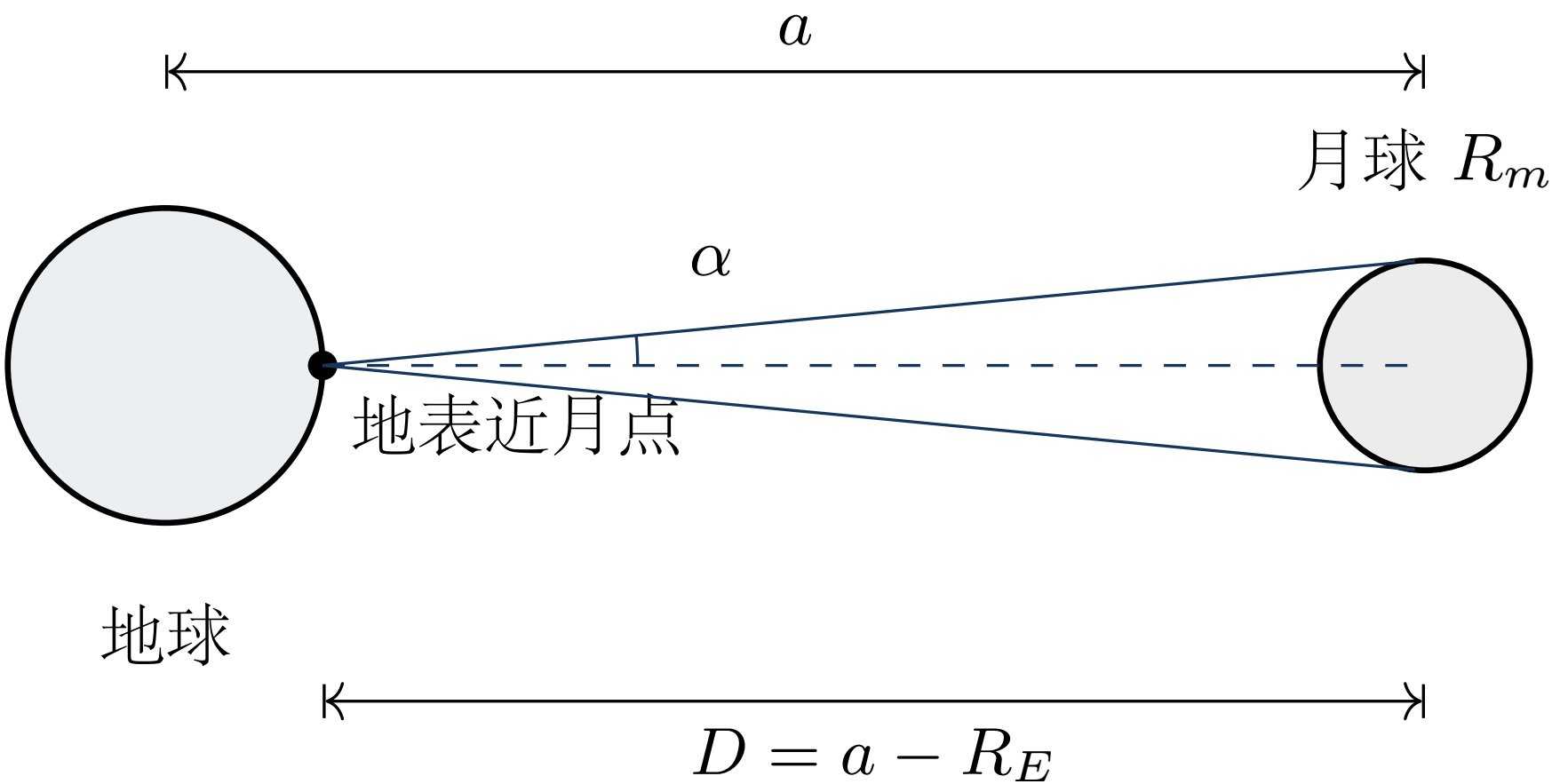

$$\mathcal{O} = 2\pi(1 - \cos\alpha) \simeq \pi R_m^2 / D^2$$